The Wald Test can be extended to testing multiple parameters simultaneously by noting the following result: If $\mathbf{X}$ is p-dimensional multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, then

$$(\mathbf{X}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu})$$

$\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2_1$

has the chi-squared distribution with p degrees of freedom.

Exercise: Explain how this result can be used in conjunction with a multi-parameter distribution when the MLE is used as the estimator.

"X" replaced by "$\hat{\boldsymbol{\theta}}$", "$\mu$" replaced with $\boldsymbol{\theta}_0$

Σ is $\mathbf{I}^{-1}(\theta_0)/n$, i.e. $\boldsymbol{\Sigma}^{-1} = n\mathbf{I}(\theta_0)$

Under $H_0: \boldsymbol{\theta} = \boldsymbol{\theta}_0$, $T = (\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)^T n\mathbf{I}(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0) \sim \chi^2_p$.

where p = # of params in $\boldsymbol{\theta}$, $H_1: \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$

reject H_0 if T is large, p-value $= P(T \geq T_{obs} \mid T \sim \chi^2_p)$

Other Tests

A wide range of tests appear in various situations. They typically share the same framework: For some choice of test statistic, the distribution of that statistic is known (approximately, at least) under the null hypothesis. From this, a rejection region can be constructed. Most software packages report p-values for tests.

For example, there are tests for normality. These are hypothesis tests whose **null hypothesis** is that the given sample was drawn from a normal distribution. As appealing as this sounds, this can be tricky. A distribution has many features, and boiling those down to a single number (the test statistic) is not easy.

Exercise: Why is it important to keep in mind that the null hypothesis of a normality test is that the sample is drawn from a normal distribution?

The test cannot find evidence that the sample was drawn from a normal dist.

Only can find if there is strong evidence in support of non-normality.

The Jarque-Bera Test

One example of a test for normality is the Jarque-Bera Test.

The test statistic in this case compares the skewness (S) and kurtosis (K) of the sample to what is expected under a normal distribution.

Skewness is a measure of how far from symmetric a distribution is:

$$\text{skewness} = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$

while kurtosis quantifies the amount of probability in the tails relative to the "center" of the distribution:

$$\text{kurtosis} = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right]$$

Normal distributions have skewness of zero and kurtosis of three. So, Jarque and Bera constructed a test statistic as follows:

$$T = n\left(\frac{S^2}{6} + \frac{(K-3)^2}{24}\right)$$

Under the null hypothesis that the sample is drawn from a normal population, T is approximately chi-squared distributed with two degrees of freedom.

In R, this is implemented as `jarque.bera.test()` in package `tseries`:

```
> library(tseries)
> jarque.bera.test(rnorm(100))

    Jarque Bera Test

data:  rnorm(100)
X-squared = 1.7944, df = 2, p-value = 0.4077
```


Exercise: Assess the performance of the test when $n = 100$ with $\alpha = 0.05$. Suppose that the true distribution is t with five degrees of freedom. What proportion of the time does the test reject the null?

Note that over 1/3 of the time the test fails to reject.

Again, we see that "fail to reject Ho" should not imply "strong evidence in favor of Ho"

STREETWISE | By James Mackintosh

Kicking the Tires Of Market Anomalies

Tie together an algorithm, an exchange-traded fund and an academic study finding an anomaly in the markets, and voilà! You have a formula for making money. The trouble is, it turns out that most of the supposed anomalies academics have identified don't exist, or are too small to matter.

A new study making waves in quantitative finance tested 447 anomalies identified by academics and found more than eight out of 10 vanish when rigorous tests are applied. Among those failing to reach statistical significance: one anomaly recently set out by the godfathers of quantitative finance, Nobel-winning economist Eugene Fama and his colleague Kenneth French.

The study, "Replicating Anomalies," published this week by Kewei Hou and Lu Zhang at Ohio State University and Chen Xue at the University of Cincinnati, is the biggest test of examples of inefficient markets carried out so far. The trio applied consistent analysis to the supposed anomalies, used the same database of stocks and set higher standards for statistical significance. Simply reducing the influence of the plethora of rarely traded penny stocks—which make up just 3% of market value but 60% of all listings—by using market capitalization weightings made more than half of past findings no longer significant.

Messrs. Hou, Xue and Zhang warn that academics have been fiddling the statistics to come up with interesting findings, known to statisticians as data mining or p-hacking. "The anomalies literature is infested with widespread p-hacking," they write.

It isn't all bad news for investors and those trying to make a living flogging what have become known as "factors." The research confirmed that the most popular factors have indeed outperformed the market over long periods even when faced with rigorous tests, but found much smaller returns than previous studies estimated.

Market anomalies that passed the new study's tests included several of the biggest. Cheap stocks indeed beat expensive ones; share prices have momentum; companies that invest a lot underperform, and quality of

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STREET

Continued from the prior page

earnings matters. Known as value, momentum, investment and quality, these have become the biggest of the so-called "smart beta" ETFs sucking in tens of billions of dollars.

A lot depends on exactly how the factors are implemented, though, and the researchers dismissed one of the industry-standard Fama-French factors as statistically insignificant: Companies with high operating return on equity don't outperform meaningfully on their tests. Other measures of return on equity did outperform sufficiently, however, underlining the sensitivity of some factors to exactly how they are defined.

One lesson for investors is to be careful about trying to make money by repeating what seems to have worked in the past. If it was so easy, everyone would do it and it would stop working.

A former student of Mr. Fama, Cliff Asness, founder of quantitative hedge-fund manager AQR Capital Management, said he tries to avoid being caught out by false findings by trading on anomalies he can explain, economically or through investor behavior. To assess whether the market anomalies will continue, he looks for ones which carried on after being identified, can be seen in other markets or asset classes, and where minor changes to how they are defined don't much affect the result. These include most famously value, momentum and corporate quality, among others.

Still, he worries that the "awesome effort" in the new paper might lead some to overreact and reject all factors, even those which Messrs. Hou, Xue and Zhang found evidence for.

"Many factors are demonstrably silly, or are highly correlated versions of the same idea," he said. "Where I get worried is about overreaction [to the paper] and the cynicism it breeds."

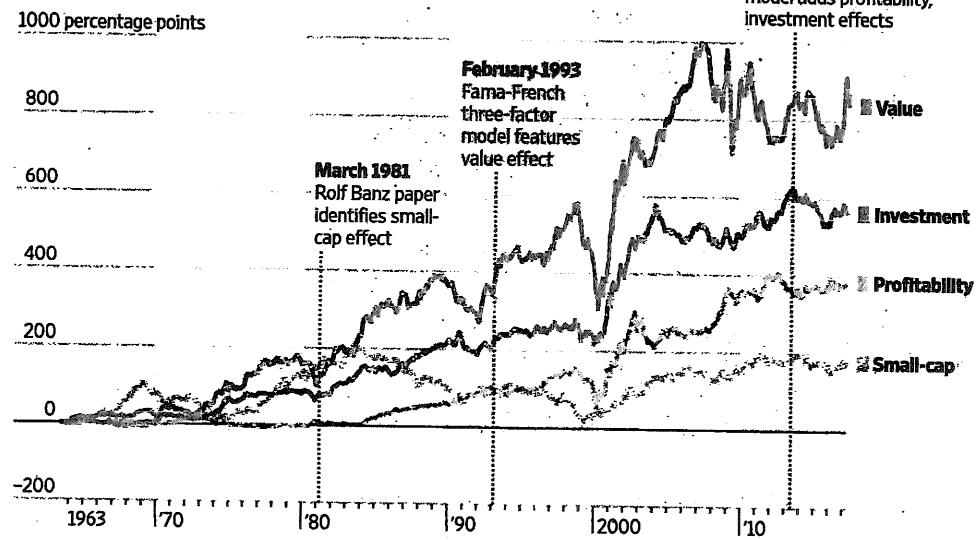
Investors are still likely to be confused. There are well over 100 value and high-dividend ETFs in the U.S. alone, tracking large, small or mid-size stocks, based on different definitions and often combined with other factors such as momentum, quality or low volatility. Intelligently choosing between them would mean examining how indexes are constructed and comparing to the long-term academic studies to see which methodology was best; in practice, for most investors, there is little more to go on than a few years of performance data and fees.

Messrs. Hou, Xue and Zhang provide a handy dismissal of factors that didn't even work that well in the past. But ultimately no one knows whether even previously robust factors like value and momentum will keep working.

Factors to Beat the U.S. Market

Academics often identify stock anomalies that have beaten the market in the past. Many were simply wrong, according to a new study, but some stopped working after publication and others turned out to be weaker than thought.

Factor cumulative outperformance*



*Performance of long-short portfolios on each factor. Investment factor is long companies which invest little, short companies which invest a lot. Note: Through March 31
Source: Kenneth French

THE WALL STREET JOURNAL.

The New York Times Magazine https://nyti.ms/2k2HYki

Magazine | ON MEDICINE

A Failure to Heal

By SIDDHARTHA MUKHERJEE NOV. 28, 2017

What happens when a clinical trial fails? This year, the Food and Drug Administration approved some 40 new medicines to treat human illnesses, including 13 for cancer, three for heart and blood diseases and one for Parkinson's. We can argue about which of these drugs represent transformative advances (a new medicine for breast cancer, tested on women with relapsed or refractory disease, increased survival by just a few months; a drug for a type of leukemia had a more lasting impact), but we know, roughly, the chain of events that unfolds when a trial is positive. The drug is approved for human use; "postapproval marketing" is deployed to commercialize the treatments; slick ads materialize on TV; fortunes are built. Yet the vastly more common experience in the life of a clinical scientist is failure: A pivotal trial does not meet its expected outcome. What happens then?

A few years ago, I was a lead investigator in a study for a drug for blood cancer. Let's call the medicine O. The compound, designed to kill leukemia cells, had shown efficacy on cancer cells in Petri dishes. The trial was backed by a small company with just a handful of employees, many of whom had invested their lives, and their life savings, in the company.

The first patient to enroll was a 60-something woman whom my colleagues and I had been treating with other medicines. Unfortunately, all the other drugs had stopped working. Her illness caused her bone marrow — the body's nursery for the genesis of blood cells — to malfunction, and so her blood counts would collapse every two weeks. She would be back at the hospital, awaiting a blood transfusion. The repeated transfusions, in turn, provoked immune responses, making it hard to find a match for her. She would wait for hours, or even days, before we could find a subtype of blood that would not be rejected by her body.

We started the experimental medicine on a Monday. When she returned to the clinic two weeks after, I could see her face illuminated with the warm flush of color that is instantly recognizable to a hematologist. Her blood counts were up. We bumped fists for the first time in our lives — she was usually more formal — and I sent her home with a congratulatory nod. But then the response flickered off. Her blood counts sank again. A few weeks passed, and the drug stopped working altogether.

In 74 hospitals, where other trials were ongoing, patients were also struggling with strange, flickering responses. From 2003 to 2006, nearly 300 patients were enrolled in a randomized trial — with one group receiving the medicine and the other a placebo. The overall survival rates showed no difference.

In his book "Awakenings," Oliver Sacks wrote about a group of patients who, having had encephalitis decades before, had been locked into a near-paralyzed catatonia. In the spring of 1969, Sacks began to treat them with a drug called L-dopa. The patients, miraculously, "awoke" from their locked-in states. Some walked the corridors; some began to speak, recounting stories of lives that had been deep-frozen in a neurological tundra. It was a coming-to-life moment that Sacks described in the book's prologue as "the most significant and extraordinary in my life." He was "caught up with the emotion, the excitement, and with something akin to enchantment, even awe."

But missing from the prologue is another story. The awakening was temporary. One by one, the responses foundered and died: Nearly all the patients became resistant and returned to their catatonic states. For me as a young clinical scientist, it was not the awakening but its opposite — the deadening — that remained as the lasting impression of the book.

The first thing you feel when a trial fails is a sense of shame. You've let your patients down. You know, of course, that experimental drugs have a poor track record — but even so, *this* drug had seemed so promising (you cannot erase the image of the cancer cells dying under the microscope). You feel as if you've shortchanged the Hippocratic oath. You think of the woman who rode the subway from Queens and spent three hours in the ice box of a waiting room on 166th Street to get her intravenous infusion. She might not have suffered bodily damage — but you've indubitably done her some harm.

There's also a more existential shame. In an era when Big Pharma might have macerated the last drips of wonder out of us, it's worth reiterating the fact: Medicines are notoriously hard to discover. The cosmos yields human drugs rarely and begrudgingly — and when a promising candidate fails to work, it is as if yet another chemical morsel of the universe has been thrown into the Dumpster. The meniscus of disappointment rises inside you: That domain of human biology that the medicine hoped to target may never be breached therapeutically. Of the several million chemical reactions in the human body, one estimate suggests that only 250 — a fraction of a percent — are currently targeted by our pharmacopoeia (this number changes every year, of course). The rest of our physiology is still impenetrable — invisible to pharmacology, like dark matter.

And then a second instinct takes over: Why not try to find the people for whom the drug *did* work? In O's case, the sickest patients in the study had, indeed, developed a response. Couldn't we justify using O for these patients?

This kind of search-and-rescue mission is called "post hoc" analysis. It's exhilarating — and dangerous. On one hand, it promises the possibility of resuscitating the medicine: Find the right group of responsive patients within the trial group — men above 60, say, or postmenopausal women — and you can, perhaps, pull the drug out of the rubble of the failed study.

But it's also a treacherous seduction. The reasoning is fatally circular — a just-so story. You go hunting for groups of patients that happened to respond — and then you turn around and claim that the drug "worked" on, um, those very patients that you found. (It's quite different if the subgroups are defined *before* the trial. There's still the statistical danger of overparsing the groups, but the reasoning is fundamentally less circular.) It would be as if Sacks, having found that the three long-term responders to L-dopa happened to be 80-year-old women from one nursing home, then published a study claiming that the drug "worked" on Brooklyn octogenarians.

Perhaps the most stinging reminder of these pitfalls comes from a timeless paper published by the statistician Richard Peto. In 1988, Peto and colleagues had finished an enormous randomized trial on 17,000 patients that proved the benefit of aspirin after a heart attack. The Lancet agreed to publish the data, but with a catch: The editors wanted to determine which patients had benefited the most. Older or younger subjects? Men or women?

Peto, a statistical rigorist, refused — such analyses would inevitably lead to artifactual conclusions — but the editors persisted, declining to advance the paper otherwise. Peto sent the paper back, but with a prank buried inside. The clinical subgroups were there, as requested — but he had inserted an additional one: "The patients were subdivided into 12 ... groups according to their medieval astrological birth signs." When the tongue-in-cheek zodiac subgroups were analyzed, Geminis and Libras were found to have no benefit from aspirin, but the drug "produced halving of risk if you were born under Capricorn." Peto now insisted that the "astrological subgroups" also be included in the paper — in part to serve as a moral lesson for posterity. I've often thought of Peto's paper as required reading for every medical student.

Why do we do it then? Why do we persist in parsing a dead study — "data dredging," as it's pejoratively known? One answer — unpleasant but real — is that pharmaceutical companies want to put a positive spin on their drugs, even when the trials fail to show benefit. ("But within a subpopulation of subjects, the results were positive." The F.D.A., though, does not approve drugs based on post hoc data.)

The less cynical answer is that we genuinely want to understand why a medicine doesn't work. Perhaps, we reason, the analysis will yield an insight on how to mount a second study — this time focusing the treatment on, say, just men over 60 who carry a genetic marker. We try to make sense of the biology: Maybe the drug was uniquely metabolized in those men, or maybe some physiological feature of elderly patients made them particularly susceptible.

Occasionally, this dredging will indeed lead to a successful follow-up trial (in the case of O, there's now a new study focused on the sickest patients). But sometimes, as Peto reminds us, we'll end up chasing mirages — trying to discover biology but ending up with astrology. When a trial fails, a clinical scientist's life thus enters a new kind of limbo. It's like being suspended somewhere between the possibility of a real awakening and the dread of loss.

Siddhartha Mukherjee is the author of "The Emperor of All Maladies: A Biography of Cancer" and, more recently, "The Gene: An Intimate History."

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A version of this article appears in print on December 3, 2017, on Page MM14 of the Sunday Magazine with the headline: When a clinical trial falters, doctors find themselves sifting through the rubble.

The Multiple Testing Problem

Possibly the biggest drawback of this hypothesis testing framework is the multiple testing problem.

Exercise: Suppose a sequence of 100 independent hypothesis tests are conducted, in every case the null hypothesis is true, and $\alpha = 0.05$ is used. How many of these tests will reject the null?

# of rejections ~ binomial($n = 100$, $p = 0.05$)

E(# of rejections) = 5

Exercise: Suppose that m independent hypothesis tests are conducted, in every case the null hypothesis is true. What is the probability of at least one rejection, if the same α is used for all tests?

P(at least 1 rejection) = 1 − P(no rejections)

$= 1 - (1-\alpha)^m$

when $m = 10$, $\alpha = 0.05$, P(at least 1 reject) = 0.40

P(at least one Type I error) = 0.40 >> 0.05 = α

In recognition of the multiple testing problem, recent advances have been made in controlling the false discovery rate (FDR), instead of worrying about making one or more Type I error.

The false discovery rate can be defined as follows:

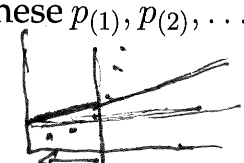
$$\text{FDR} = \frac{\text{number of false positives}}{\text{number of rejections}}.$$

Hence, FDR is the proportion of the rejected null hypotheses which are "false discoveries."

We should be clear: This approach is "solving" the multiple testing problem by, more or less, changing the objective. Still, FDR seems like a very reasonable alternative to control instead of limiting the probability of a Type I error.

Benjamini and Hochberg (*JRSSB*, Vol. 57) presented a simple procedure which limits the expected value of the FDR below a chosen α level:

1. Sort the p-values from smallest to largest, denote these $p_{(1)}, p_{(2)}, \ldots, p_{(m)}$.
2. Find the largest k such that $p_{(k)} \leq k\alpha/m$



3. The hypotheses with p-values less than or equal to $p_{(k)}$ can be claimed as "discoveries" for this choice of α.

It should be stressed that the validity of this procedure assumes that the individual hypothesis tests are **independent** or satisfy certain dependency conditions. Extensions do exist; for example, the Benjamini-Hochberg-Yekutielli procedure divides the factor $k\alpha/m$ by $\Sigma_{i=1}^{m} 1/i$ and works under a wider range of **positive** dependence situations.

The function `p.adjust()` in R, when used with argument `method="BH"` will adjust the supplied p-values, returning the smallest choice of α such that a discovery would be claimed for each test.

Interesting Fact: Assuming the null hypothesis is true, the p-value is a random variable with the Uniform$(0, 1)$ distribution.

```
> simpvals = runif(1000)
> print(mean(simpvals < 0.05))
[1] 0.049
```

Simulated p-values for 1000 tests where H_0 is true in all cases

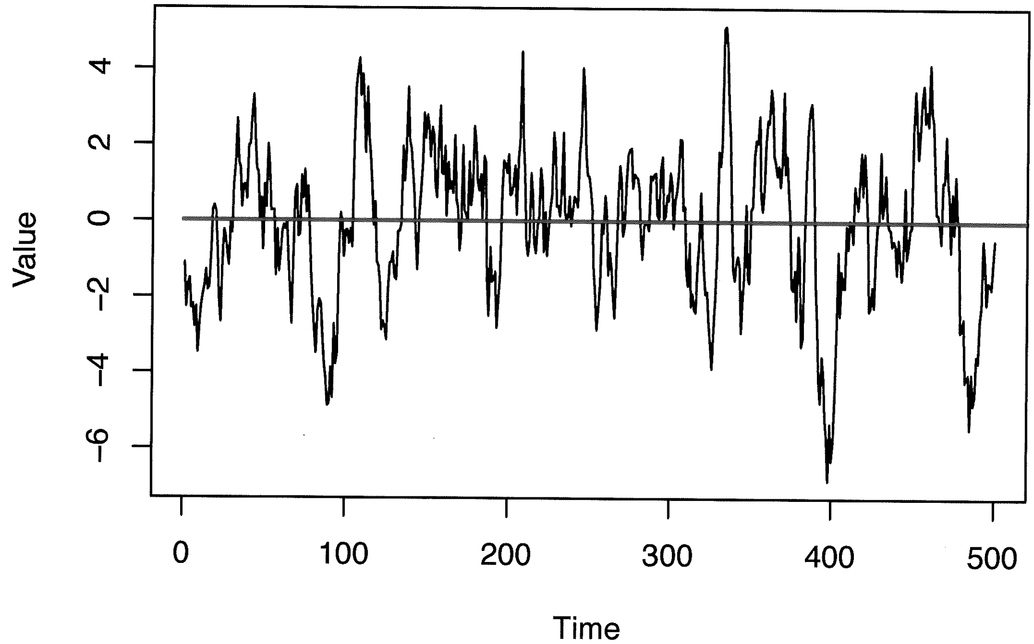
```
> print(mean(p.adjust(simpvals, method="BH")<0.05))
[1] 0

> print(mean(p.adjust(simpvals, method="BH")<0.5))
[1] 0
```


Example: Simple Pairs Trading

A time series is a sequence of random variables (or, the actual observed values of those random variables) that model a sequence of data over time. Typically these observations are assumed to be equally spaced in time. Denote this $X_1, X_2, \ldots$ or $\{X_t\}$.

A time series is said to be stationary if the **uncondtional** mean and variance is constant ($E(X_i) = \mu$ and $V(X_i) = \sigma^2$) and the covariance structure also does not change with time, i.e., $\text{Cov}(X_t, X_{t+h})$ only depends on h, and not on t, for any choice of h and t. Stationarity is a crucial concept for time series models, and can be thought of as a more realistic assumption than independence.

Stationary time series exhibit mean reversion: The series may move away from μ for periods of time, but it will inevitably drift back to μ. The example below shows a series simulated from the AR(1) model, with $\mu = 0$.



Exercise: Can time series of stock prices be modelled using stationary time series models?

No, because mean reversion would lead to easy way to make money

A pair of time series $\{X_t\}$ and $\{Y_t\}$ are said to be cointegrated if there exists a constant λ such that $X_t - \lambda Y_t$ is stationary. Note that this does not require that either $\{X_t\}$ or $\{Y_t\}$ be stationary.

This sets up a simple pairs trading strategy: **If** one can finds pairs of stocks whose prices are cointegrated, then we can use mean reversion of $X_t - \lambda Y_t$ to execute trades that will reliably make money.

Hypothesis tests do exist that test for cointegration. Here, we will utilize the Phillips-Ouliaris Cointegration Test, implemented as function `po.test()` in package `tseries`.

Exercise: What is the null hypothesis with this cointegration test? Does that seem to be the reasonable choice?

H_0: the series are <u>not</u> cointegrated

H_1: the series are cointegrated

Important that it be set up in this way so that we can claim evidence of cointegration

Exercise: One could search over many pairs of stocks in an attempt to find cointegrated pairs. What are potential problems with this approach?

A massive multiple testing problem.

Start with m stocks, there are $\binom{m}{2}$ possible pairs. A lot of tests.

Recall that for our discussion of dimension reduction we constructed a sample of 1000 NYSE stocks, each with 30 days of observed prices. We will use these data to search for cointegrated pairs.

First, read in the sample and randomly pair up the 1000 symbols:

```
> stocksample = read.table("stocksample.txt", header=T,
+                          sep="\t", comment.char="")
> set.seed(0)
> sp = matrix(sample(1:1000,size=1000,replace=F),ncol=2)
```

Now calculate the 500 p-values for each of the tests:

```
> pval = numeric(nrow(sp))
> for(i in 1:nrow(sp))
+ {
+  testout = po.test(log(t(as.matrix(stocksample[sp[i,],5:34]))))
+  pval[i] = testout$p.value
+ }
```

How many of the p-values are less than 0.05?

```
> sum(pval < 0.05)
[1] 17
```

What happens when we try to control the false discovery rate?

```
> sum(p.adjust(pval, method="BH") < 0.05)
[1] 0
```

This is a perfect example of why one needs to be careful of the multiple testing problem. If you search over enough pairs, some are going to appear to be cointegrated. False positives are inevitable.

Part 16: Bayesian Estimation

Summary

To this point, our discussion has focused on the frequentist or classical approach to estimation and hypothesis testing. In that framework, consideration is given to how procedures perform in repeated use. Indeed, the performance of estimators can be assessed without using any actual data. Some find this troubling.

An alternative philosophy is the Bayesian approach to estimation. The central characteristic is that uncertainty in parameters is quantified using probability distributions.

As the name implies, Bayes' Rule plays a central role.

First, recall Bayes' Rule: If $B_1, B_2, \ldots, B_m$ are a partition of Ω, then

$$P(B_i|A) = P(A|B_i)P(B_i) \Big/ \sum_j P(A|B_j)P(B_j)$$

Also recall the continuous version:

$$f_{Y|X}(y|x) = f_{X|Y}(x|y)f_Y(y) \Big/ \int f_{X|Y}(x|u)f_Y(u)\,du$$

Here we will use this result with the parameter θ "playing the role" of Y:

$$\pi(\theta|\mathbf{x}) = f(\mathbf{x}|\theta)\pi(\theta) \Big/ \int f(\mathbf{x}|t)\pi(t)\,dt$$

The vector $\mathbf{x}$ is observed data.

The density $\pi(\theta)$ is the prior distribution for θ.

The conditional density $\pi(\theta|\mathbf{x})$ is the posterior distribution for θ.

In the Bayesian framework, the prior reflects the someone's **opinion** regarding the value of θ, **prior to** the observation of the data.

The posterior is this opinion "updated" to take the data into consideration.

How is the prior chosen? This can be tricky, and subjective.

The use of the Bayesian approach was long limited by computational difficulties associated with calculating the required integral. With the advent of Markov Chain Monte Carlo (MCMC) algorithms, these issues have largely gone away, and the use of Bayesian methods has grown greatly.

In some relatively simple cases, analytical solutions are possible.

Exercise: Suppose that the observable data are modelled by i.i.d. Bernoulli(θ) random variables $X_1, X_2, \ldots, X_n$. You quantify your prior beliefs regarding θ by using the Beta(α, β) distribution, for some chosen values of α and β. Derive the posterior for θ in this case.

First, recall the Beta dist:

$$\pi(\theta) = \left(\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\right) \theta^{\alpha-1} (1-\theta)^{\beta-1} \quad 0<\theta<1$$

The dist for data $f(\underset{\sim}{X} \mid \theta)$

$$f(\underset{\sim}{X} \mid \theta) = \prod_{i=1}^{n} f(X_i \mid \theta) = \prod_{i=1}^{n} \theta^{X_i} (1-\theta)^{1-X_i}$$

$$= \theta^{\sum X_i} (1-\theta)^{n-\sum X_i}$$

Now use Bayes Rule

$$\pi(\theta \mid \underset{\sim}{x}) = \frac{f(\underset{\sim}{x} \mid \theta)\,\pi(\theta)}{\int f(\underset{\sim}{x} \mid t)\,\pi(t)\,dt} = k_1 f(\underset{\sim}{x} \mid \theta)\,\pi(\theta)$$

where k_1 is not a function of θ

$$= \left[\theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\right]\left[\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\,\theta^{\alpha-1}(1-\theta)^{\beta-1}\right]$$

$$= k_2\,\theta^{(\alpha-1)+\sum x_i}(1-\theta)^{(\beta-1)+(n-\sum x_i)}, \quad 0<\theta<1$$

where k_2 does not depend on θ

So, that this is the Beta$\left(\alpha+\sum x_i,\ \beta+(n-\sum x_i)\right)$ distribution.

In this example, we see that the prior and posterior are from the same family of distributions (i.e., they are both Beta). We say that "The Beta distribution is conjugate to sampling from the Bernoulli."

Let's consider a couple specific examples. Suppose I chose $\alpha = 2$ and $\beta = 18$ to capture my prior opinion regarding θ. I observe $n = 5$ trials, and $\sum_i x_i = 1$. The result above states that the posterior is $\text{Beta}(2+1, 18+(5-1))$.

If instead, $n = 500$ and $\sum_i = 100$, the posterior is $\text{Beta}(102, 418)$.

The figures on the following slide compare these cases.